Arin Wongprommoon

arin@tropmedres.ac
arinwongprommoon.github.io (personal website) • linkedin.com/in/arinwongprommoon (LinkedIn profile) • 0000-0001-9316-8336 (ORCID) • github.com/arinwongprommoon (code repository)
Work address: 8/F Santasiri Sornmani Building, Mahidol-Oxford Tropical Medicine Research Unit, Faculty of Tropical Medicine, Mahidol University, 420/6 Ratchawithi Road, Ratchathewi District, Bangkok 10400, Thailand

---

I am committed to improving health outcomes in socioeconomically disadvantaged and underrepresented populations by using data-intensive methods that identify clinical and biological markers of disease. Using my broad experience and skills in computational biology and data science, I investigate host-pathogen interactions in melioidosis — a neglected tropical disease caused by a bacterium and endemic to Thailand — that may inform patient stratification and therapies. My future goal is to lead an investigation into creating a tool to integrate multi-modal omics data to inform a mathematical model that predicts risk of death from melioidosis based on host and pathogen determinants.
Alongside my scientific work, I build new research capacity through mentoring interdisciplinary and international collaborative projects. I am also committed to improving the lives of those belonging to marginalised groups through shaping policies in Equality, Diversity, and Inclusion at my home organisations.

Education

Oct 2019 – Jan 2024
- **PhD in Quantitative Biology, Biochemistry, and Biotechnology**, University of Edinburgh, United Kingdom.
  Thesis title: *Single-cell time-series analysis of metabolic rhythms in yeast*
  Funded by Edinburgh Global Scholarship and School of Biological Sciences, University of Edinburgh.

Oct 2016 – Jun 2019
- **BA Hons Natural Sciences (Biochemistry)**, University of Cambridge, United Kingdom.
  Final-year research project title: *Sequence Preferences of the Nucleosome and PCR Enzymes*
  Funded by King's Scholarship (Royal Thai Government).

Employment

Jan 2024 – present
- **Postdoctoral Fellow,** Joint appointment between Mahidol-Oxford Tropical Medicine Research Unit (MORU), Bangkok, Thailand & Wellcome Sanger Institute, Hinxton, United Kingdom.
  I combine bacterial genomics data and human transcriptomics data to explain the variability of outcomes of melioidosis, a neglected tropical disease caused by the bacterium *Burkholderia pseudomallei* and train and supervise trainees working with the research group. I have identified key alternative splicing patterns that were enriched in fatal melioidosis patients from a cohort in northeast Thailand and have combined paired host-pathogen data from five studies to inform a bacterial genome-wide association study.

Publications

1. Kronsteiner, B., Twumasi, C., Abraham, P., Chumseng, S., Chaichana, P., Boonklang, P., **Wongprommoon, A.,** Angchagun, K., Chantratita, N., Limmathurotsakul, D., Day, N. P. J., Chamnan, P., Chewapreecha, C. & Dunachie, S. J. Metformin use is associated with lower mortality from bacterial sepsis and improved immunocompetence in Thai diabetes patients with acute melioidosis. *medRxiv,* 2025.10.29.25338967. doi:`10.1101/2025.10.29.25338967` (2025).
2. Lázaro, J., **Wongprommoon, A.,** Júlvez, J. & Oliver, S. G. Enhancing Genome-Scale Metabolic Models with Kinetic Data: Resolving Growth and Citramalate Production Trade-offs in E. coli. *Bioinformatics Advances,* vbaf166. doi:`10.1093/bioadv/vbaf166` (2025).
3. Pakdeerat, S., Chomkatekaew, C., Boonklang, P., **Wongprommoon, A.,** Angchagun, K., Dokket, Y., Faosap, A., Wongsuwan, G., Amornchai, P., Wuthiekanun, V., Changklom, J., Siriboon, S., Chamnan, P., Peacock, S. J., Parkhill, J., Corander, J., Day, N. P., Thomson, N. R., Uttamapinant, C., Wongpalee, S. P. & Chewapreecha, C. CRISPR-based environmental detection of Burkholderia pseudomallei identifies sanitation gaps and melioidosis risk in northeast Thailand. *medRxiv* (*under review with Nature Communications*), 2024.11.21.24317607. doi:`10.1101/2024.11.21.24317607` (2025).
4. **Wongprommoon, A.,** Muñoz, A. F., Oyarzún, D. A. & Swain, P. S. Single-Cell Metabolic Oscillations Are Pervasive and May Alleviate a Proteome Constraint. *bioRxiv* (*under review with Nature Communications*), 2024.11.25.625147. doi:`10.1101/2024.11.25.625147` (2024).
5. **Wongprommoon, A.,** Chomkatekaew, C. & Chewapreecha, C. Monitoring Pathogens in Wastewater. *Nature Reviews Microbiology,* 1–1. doi:`10.1038/s41579-024-01033-1` (2024).

6. Nikolados, E.-M., **Wongprommoon, A.,** Aodha, O. M., Cambray, G. & Oyarzún, D. A. Accuracy and Data Efficiency in Deep Learning Models of Protein Expression. *Nature Communications* **13,** 7755. doi:10.1038/s41467-022-34902-5 (2022).
7. Tipgomut, C., **Wongprommoon, A.,** Takeo, E., Ittiudomrak, T., Puthong, S. & Chanchao, C. Melittin Induced G1 Cell Cycle Arrest and Apoptosis in Chago-K1 Human Bronchogenic Carcinoma Cells and Inhibited the Differentiation of THP-1 Cells into Tumour- Associated Macrophages. *Asian Pacific journal of cancer prevention: APJCP* **19,** 3427–3434. doi:10.31557/APJCP.2018.19.12.3427. pmid: 30583665 (2018).
8. Jia, B. & **Wongprommoon, A.** Synthetic Biology: Engineering Order in Organisms across Scales and Species. *BioTechniques* **65,** 113–119. doi:10.2144/btn-2018-0121 (2018).

Conferences

Jun 2024 • **Hinxton Immunogenomics Day**, Hinxton, United Kingdom.
Oral presentation: *Effect of diabetes on host gene expression and transcript splicing responses to melioidosis in Northeast Thailand*

May 2024 • **Health Challenge Thailand**, London, United Kingdom.
Oral presentation: *Prevalence of undiagnosed diabetes and its effects on human host response to melioidosis in Northeast Thailand*

Apr 2022 • **Microbiology Society Annual Conference**, Belfast, United Kingdom.
Poster: *Metabolic cycles are robust and respond to nutrient changes in single cells of budding yeast*
Awarded Microbiology Society Grant (GBP 360), partly covering registration and travel expenses.

Dec 2021 • **British Yeast Group Meeting on *the Future of Yeast Research***, Cambridge, United Kingdom.
Poster: *Single-cell analysis shows that flavin-based yeast metabolic cycles are robust and respond to nutrient changes*
Best Poster (Graduate Student) Prize (3 recipients out of 43 posters), GBP 150.

Jun 2021 • **Cold Spring Harbor Laboratories Symposium on Quantitative Biology on *Biological Timekeeping***, Laurel Hollow, New York, United States.
Poster: *Single-cell analysis shows that flavin-based yeast metabolic cycles are robust and respond to nutrient changes*

Peer review

Jan 2024 – present • *Current Microbiology* (1 manuscript).

Research funding

Nov 2025 – Nov 2026 • MORU-OUCRU Discovery Research Academy.
Project title: *Multi-omics analysis to enable precision drug discovery for melioidosis*
Award amount: USD 30,000
Awarded funding for a seed project to combine *Burkholderia pseudomallei* GWAS with metabolic profiling of blood samples from melioidosis patients to identify host-pathogen interactions that explain mortality, with some of the budget going towards clinician engagement at a hospital in an endemic area and travel to the European Melioidosis Congress. This is part of a Wellcome-funded joint initiative to provide mentorship to early- and mid-career PhD-level researchers from south and southeast Asia (Jul 2025 – Dec 2026, 15 participants, acceptance rate 22%) on project management, writing academic CVs, project pitching, and applying for academic research grants.

Research experience

Oct 2019 – Oct 2023 • **PhD project with Biomolecular Control Group (Prof Diego Oyarzún) & Prof Peter Swain's Group, Centre for Engineering Biology, University of Edinburgh**, United Kingdom.
Showed that metabolic cycles in single *Saccharomyces cerevisiae* cells are autonomous from the cell division cycle and are persistent across nutrient conditions and gene deletions, using single-cell microfluidics and time series analysis. Part of a team to maintain an image analysis software pipeline. Predicted that biosynthesis of biomass components in sequence is a time-efficient use of limited enzyme-available proteome resources, using flux balance analysis.

Research experience (continued)

Dec 2022
- **Alan Turing Institute Data Study Group**, London, United Kingdom.
  Five-day fully-funded collaborative hackathon: worked with a team of 12 researchers and data scientists to engineer a machine learning model to predict sound annoyance of 2,980 urban sound recordings; my focus was on development infrastructure and final presentation. Found that a pre-trained audio neural network trained on high-resolution spectrograms had best prediction ability.

Jan 2019 – Mar 2019
- **Final-year undergraduate research project, under the supervision of Dr Fangjie Zhu, with Prof Jussi Taipale's group, Department of Biochemistry, University of Cambridge**, United Kingdom.
  Used nucleosome EMSA-SELEX to confirm rules for nucleosome positioning and showed that C-methylation aligns phases of dinucleotides with cytosines. Showed that enzyme-introduced biases were most responsible for PCR bias by comparing *k*-mer fold changes of sequencing libraries.

Jun 2018 – Sep 2019
- **Internship, under the supervision of Prof Jorge Júlvez, with Prof Steve Oliver's group, Cambridge Systems Biology Centre, University of Cambridge**, United Kingdom.
  Extended a kinetic model for *E. coli* metabolism to investigate reaction fluxes and enriched a stoichiometric model. Used a genetic algorithm to optimise parameters.

Jul 2017 – Sep 2017
- **Internship, with Asst Prof Maria Spletter's group, Department of Physiological Chemistry, Ludwig-Maximillian University**, Munich, Germany.
  Studied role of the splicing factor Aret in development of indirect flight muscle (IFMs) in *Drosophila melanogaster*. Found that in Aret null mutants, IFM density decreased and IFMs fused during pupal development. Poster presentation at Amgen Scholars Symposium in Cambridge won second prize among 19 LMU Munich scholars. Funded by Amgen Scholars Programme Europe.

Aug 2016 – Sep 2016
- **Internship, with Prof Chanpen Chanchao's group, Department of Biology, Chulalongkorn University**, Bangkok, Thailand.
  Studied effect of melittin (bee venom extract) on ChaGo-K-1 lung cancer and Wi-38 lung fibroblast cells. Found that IC_{50} is likely $4\,\mu g\,mL^{-1}$ to $8\,\mu g\,mL^{-1}$ and Wi-38 survived at higher melittin concentrations compared to ChaGo-K-1.

Capacity building, mentorship, and leadership

Jul 2025 – Aug 2025
- **Supervisor for intern with Dr Claire Chewapreecha's group, MORU**, Thailand.
  Supervised a first-year pharmacy undergraduate with previous programming experience during a research project. The project aimed to curate the AMRrules database of antibiotic resistance genes in *B. pseudomallei* based on the scientific literature. The student defined 127 rules, then verified the format of the new database entries using a Python script on a Unix-based operating system.

May 2025 – present
- **Executive officer for MORU Equality, Diversity, and Inclusion Committee**.
  Organised committee meetings and took minutes. Raised awareness about neurodiversity at an all-staff meeting through a poster stand.

Mar 2025
- **Organiser for *Our Relationship with Microbial Life* workshop**.
  Organised logistics (transport, catering) for collaborative visits by senior staff from the Wellcome Sanger institute and the Wellcome Trust to Siriraj Hospital; Faculty of Science, Mahidol University; and the National Institute of Health, Ministry of Public Health, all in Bangkok, Thailand. These visits preceded an international two-day workshop which inviting 40 senior researchers in genomics and microbiology to discuss the future of genomics in the microbial contexts, its ethics, and impact. During this workshop, I advised a team of illustrators on scientific content to help them summarise the event.

Capacity building, mentorship, and leadership (continued)

Jul 2024 • **Facilitator for Lightyear Foundation school visit to Wellcome Genome Campus**, United Kingdom
Received training from the Lightyear Foundation, a disabled-led charity dedicated to empowering STEM attainment for disabled young people, on how to work with disabled and neurodivergent students. Facilitated an activity for 10 students to learn scientific enquiry skills through DNA-extraction and pipetting activities. These students were at the late-primary school/early secondary school level with varied physical disabilities, learning disabilities, or neurodivergence.

Jun 2022 – Oct 2022 • **Advisor for University of Edinburgh-UHAS Ghana team**, International Genetically Engineered Machine (iGEM) competition.
Advised a nine-member undergraduate team that constructed and modelled cell-free solutions to mitigate plastic and heavy metal pollution in bodies of water in Ghana. Part of a three-advisor team; my focus was on structural modelling of proteins, documentation, and webpage development with version control. Team won gold medal (173 out of 350+ teams, scored against a rubric) and was nominated for best environmental project.

Jun 2022 – Oct 2022 • **Advisor for University of Edinburgh-UHAS Ghana team**, International Directed Evolution Competition (iDEC).
Advised a eight-member undergraduate team that engineered metallothioneins with increased silver ion binding capacity in *E. coli* using mutant screening. Part of a three-advisor team; my focus was on bioinformatics and structural modelling of proteins. Team won Best Community Building and Best Presentation Awards (out of 15+ teams)

Sep 2021 – Sep 2022 • **PhD Student Representative for Institute of Quantitative Biology**, Graduate School Staff-Student Liaison Committee.
One of two representatives of 77 PhD students in my institute among a team of 14 across the School of Biological Sciences (333 students). Proposed policies to ensure Equality, Diversity, and Inclusion and Widening Participation in a Graduate School internal review. Successfully advocated for formal recognition of student representative role. Conducted a survey of all Biological Sciences PhD students to propose measures to help with rising costs of living in Sep 2022. Organised community-building events, e.g. board games.

May 2021 – Jun 2021 • **Participant in the PhD Change Agents Programme**, University of Edinburgh
Worked with six PhD students from different social science, natural science, and engineering disciplines in a six-week policy-making and entrepreneurial skills development programme. Proposed policies for inclusive growth in communities after the COVID-19 pandemic. Presented a multimedia poster to a representative from the Institute for Public Policy Research Scotland.

Sep 2020 – Dec 2020 • **Mentor for IBioIC Leaders in Science Programme**, Edinburgh & Tranent, United Kingdom
Part of a five-mentor team. Organised weekly video-based workshops on science topics beyond the curriculum to pupils aged 15-–18. Delivered mentorship on science communication to empower pupils to teach science to younger pupils.

Jan 2019 – Nov 2019 • **Project Manager for *Evolving spatially-defined ecological interactions***, Cambridge University Synthetic Biology Society.
Supervised small student teams and encouraged them to work independently to engineer sets of physical and social interactions in *E. coli*. Verified aggregation of adhesion strains described by Glass & Riedel-Kruse (2018). In this framework, teams verified the auxotrophy of ecological strains from the Ackermann Lab (Eawag, Switzerland).

Oct 2017 – Mar 2019 • **Project Manager for *Bacterial edge detection***, Cambridge University Synthetic Biology Society.
Engineered a double genetic circuit in *E. coli* that enabled photography and edge detection, reproducing Tabor et al. (2009). The project evolved into five weekly workshops on molecular techniques in Oct–Nov 2018. 10–30 participants with biological, chemical, medical, and engineering backgrounds participated in each workshop.

Teaching

Jan 2023 – Apr 2023
- **Demonstrator & marker for *Practical Systems Biology***, University of Edinburgh.
  Masters-level course on modelling biological systems using differential equations and stochastic simulations via Python.

Sep 2022 – Dec 2022
- **Demonstrator for *Biology 1A: Variation***, University of Edinburgh.
  First-year undergraduate course on genetics and evolution; covered scientific skills, hypothesis testing, and Python.

Jan 2022 – Mar 2022
- **Demonstrator for *Genes & Gene Action 2***, University of Edinburgh.
  Second-year undergraduate course on genetics; covered basic bench biology.

Skills

Mathematics
- differential equations, stochastic simulations, graph theory, linear algebra including linear programming, signal processing (Fourier analysis, cross-correlation)
- Statistics: survival analysis, interaction analysis, linear modelling, multiple hypothesis testing, model selection
- Machine learning: clustering (modularity clustering, hierarchical clustering, k-nearest neighbours), dimensionality reduction (PCA, UMAP), building a classification model (architectures: support vector machine, random forest) and evaluating its performance (accuracy, precision/recall), feature selection, hyperparameter optimisation

Bioinformatics
- RNA splicing (`leafcutter`), differential gene expression (`edgeR`, `limma`), gene set enrichment analysis (Gene Ontology, `XGR`), pathway & transcription factor inference (`decoupler`)
- flux balance analysis & genome-scale metabolic models (`cobra`, `roadrunner`, `libsbml`)

Computing concepts
- data analysis & visualisation (Python `pandas`, `seaborn`; R `tidyverse`, `ggplot2`)
- high-performance computing & parallelisation, secure research environments, continuous integration
- workflow management (`snakemake`), object-oriented programming,

Programming languages
- Python, R, Bash, MATLAB, C

Software
- Git/GitHub/GitLab (version control & collaborative coding), Docker (virtualisation), LaTeX (typesetting), Inkscape (graphic design), UNIX-based operating systems (Linux, Arch and Ubuntu distributions; macOS)

Spoken languages
- English (IELTS 8.5/9, 2015) and Thai (native)